



Inspiring Excellence

Department of Electrical and Electronic Engineering
Hardware / Software Report Submission

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Course Title: Electrical Circuits I Laboratory

Section: 03

Experiment No: 05

Experiment Name: Verification of Thevenin's Theorem and
Maximum Power Transfer Theorem

Group No: 01

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Electrical Circuits I Laboratory

EEEB01L

Department of Electrical & Electronic Engineering (EEE)
BRAC University

Experiment No. 5

Verification of Thevenin's Theorem and Maximum Power Transfer Theorem

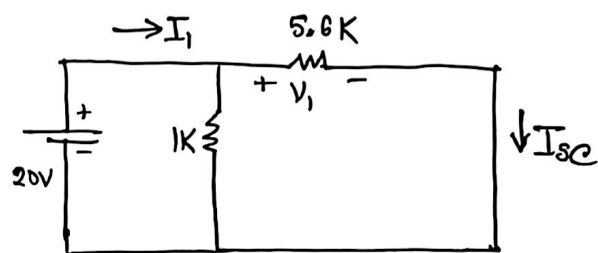
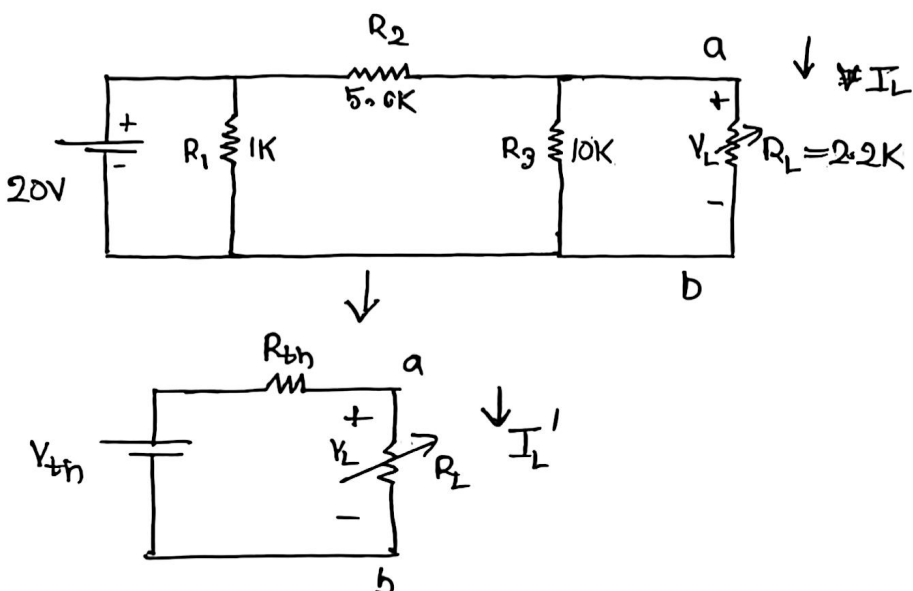
Part-A: Verification of Thevenin's Theorem

1. Objective: To verify Thevenin's theorem with reference to a given circuit.

2. Theoretical Background: Thevenin's theorem simplifies any complex, linear bilateral network into a basic equivalent circuit containing a single voltage source (V_{th}) in series with a single resistor (R_{th}) connected across the load terminals. This approach drastically minimizes repetitive calculations when analyzing the effects of changing a single branch element while the rest of the circuit remains constant.

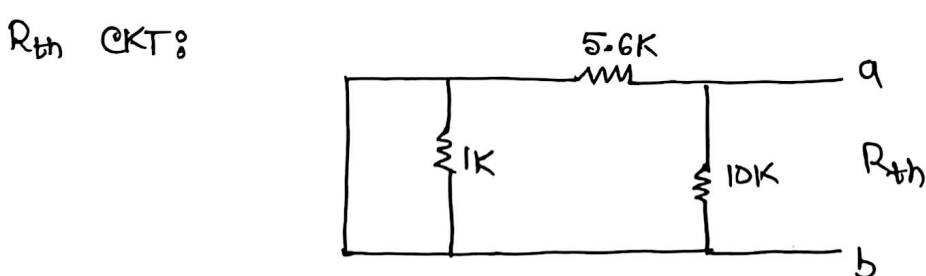
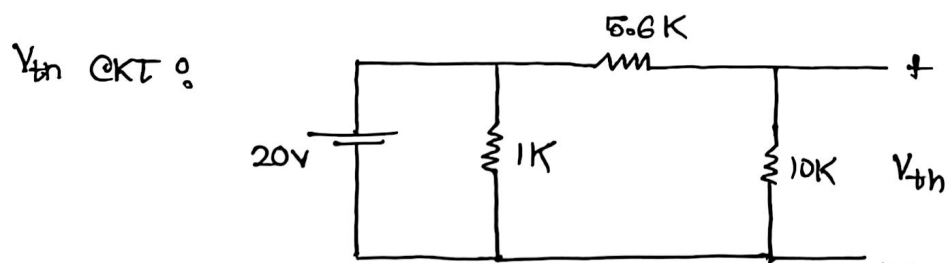
V_{th} = The open circuit voltage (V_{oc}) at two terminals A, B

R_{th} = The resistance looking into the terminals A, B of the network with all sources removed.



$$I_{sc} = I_1 = \frac{V_1}{5.6K}$$

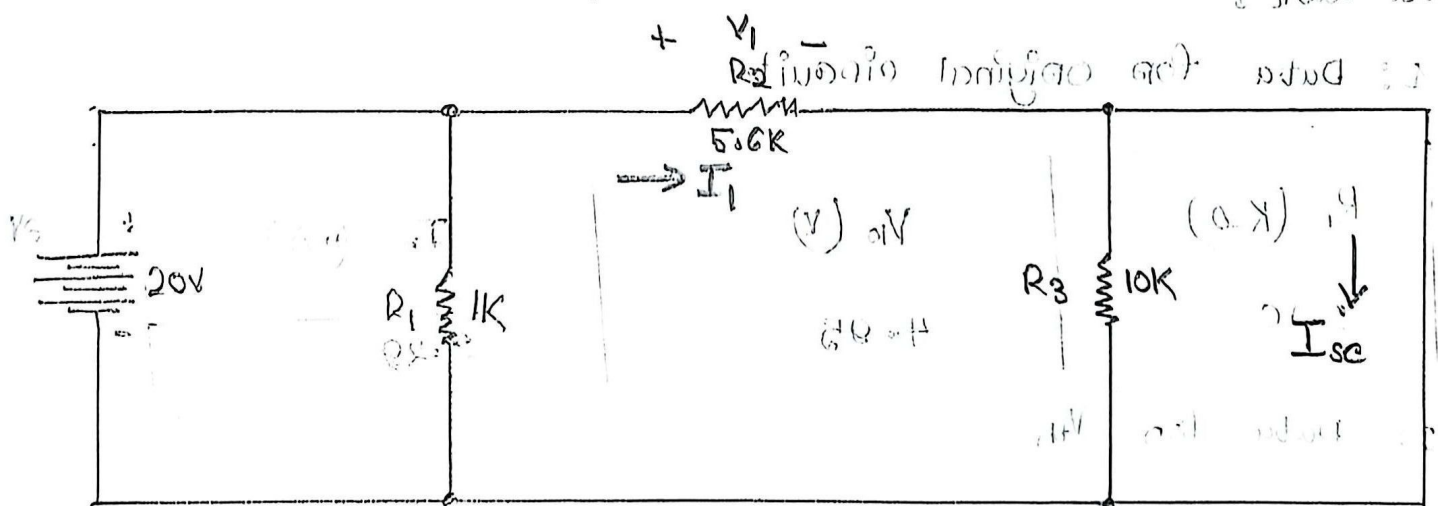
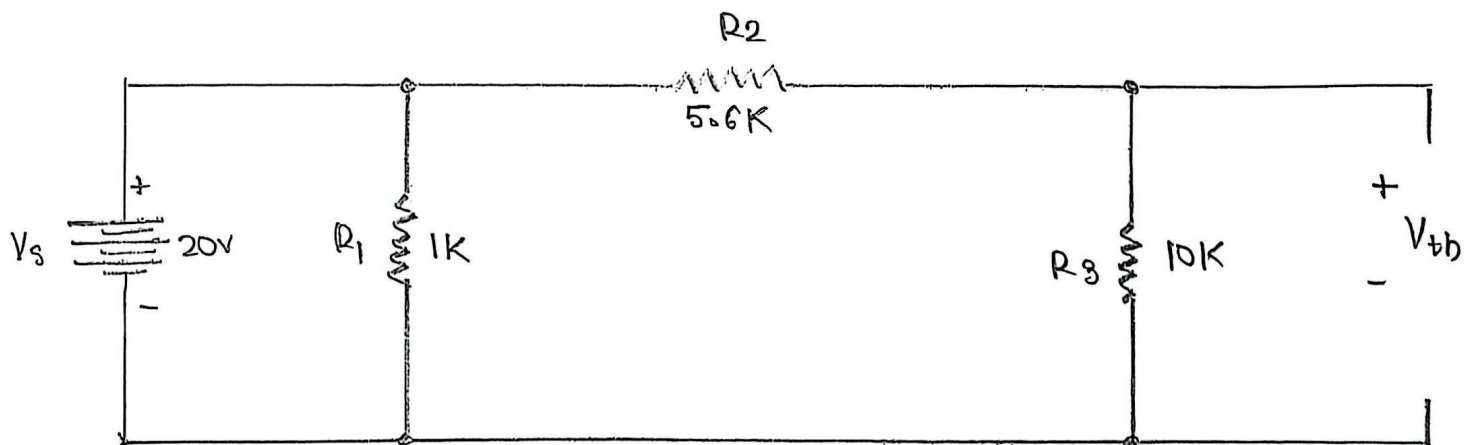
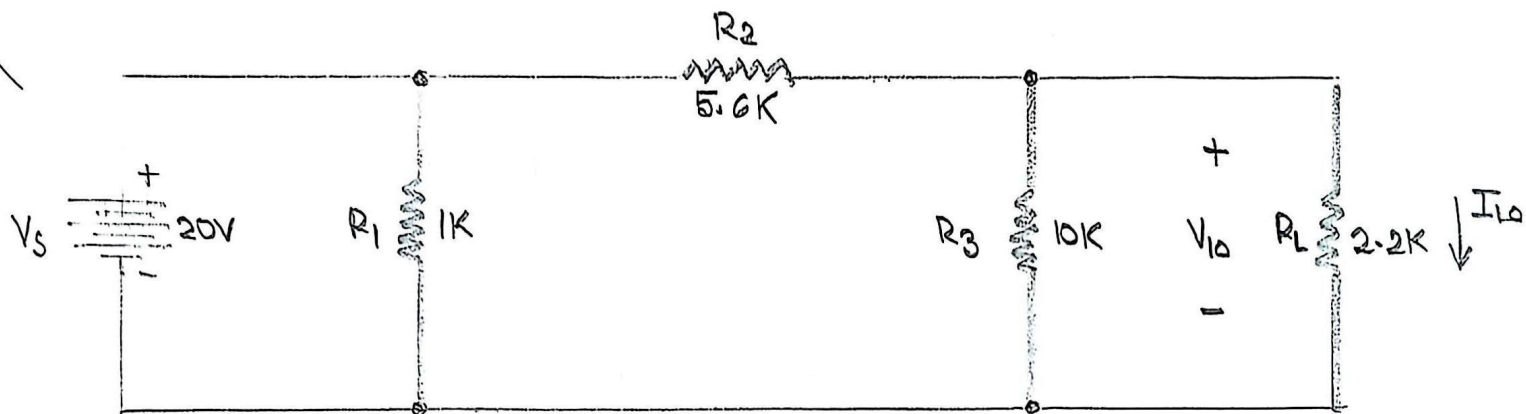
$$R_{th} = \frac{V_{th}}{I_{sc}}$$



3. Equipment:

- ⇒ Resistors
- ⇒ Multimeters
- ⇒ DC Power Supply adjusted to 20V DC
- ⇒ Breadboard

4. Circuit Diagram :



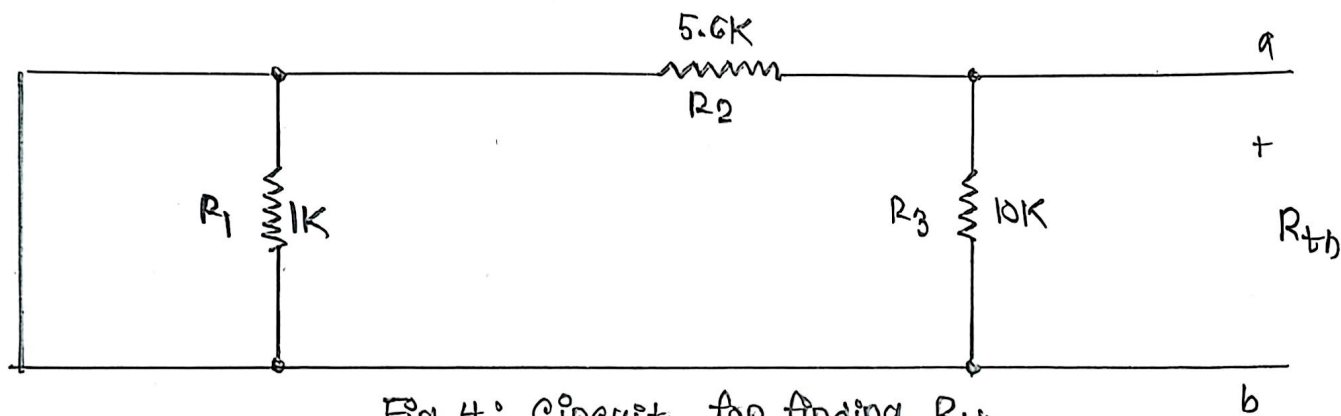


Fig 4: Circuit for finding R_{th}

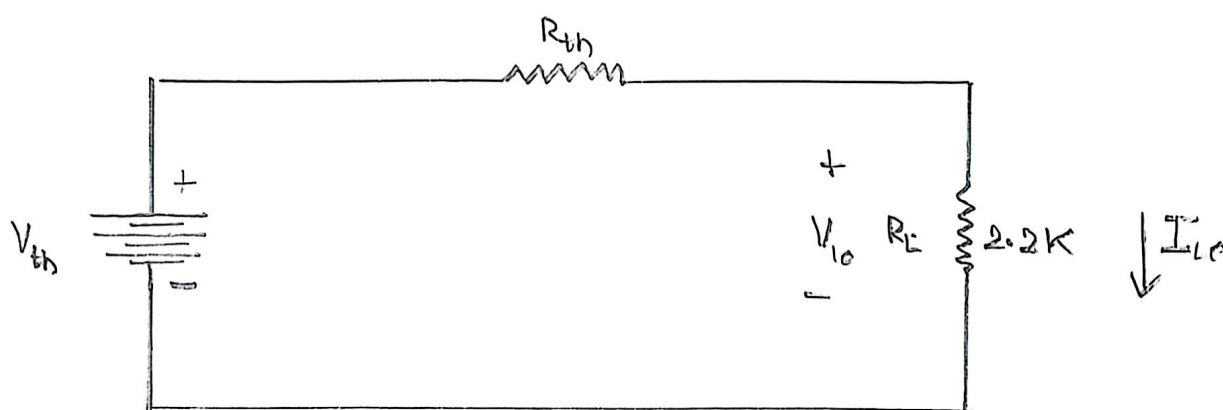


Fig 5: Equivalent circuit

5. Data Table:

Table 1: Data for original circuit

R_1 (K- Ω)	V_{lo} (V)	I_{lo} (mA)
2.16	4.95	2.29

Table 2: Data for V_{th}

V_{th} (V)
12.9

Table 3: Data for R_{th}

Indirect Method:

R_2 (K Ω)	V_1 (V)	$I_1 = V_1 / R_2$ (mA)	$I_{sc} = I_1$ (mA)	$R_{th} = V_{th} / I_{sc}$ (K Ω)
5.54	20.1	3.628	3.628	3.555

Direct Method:

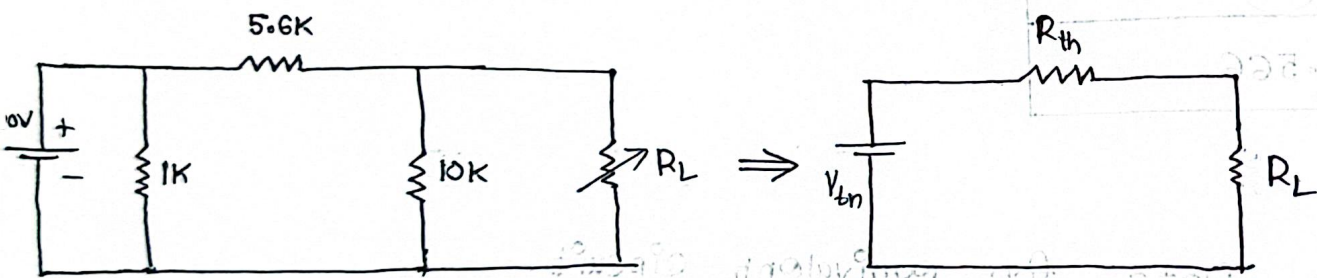
R_{th} (K Ω)
3.566

Table 4: Data for equivalent circuit

V_{ie} (V)	I_{ie} (mA)
4.95	2.25

Part-B: Maximum Power Transfer Theorem

1. Objective: The objective of this experiment is to verify maximum power transfer theorem.
2. Theoretical Background: The maximum power transfer theorem states that a resistive load will receive maximum power when its total resistive value is exactly equal to the Thevenin's resistance of the network as seen by the load.



According to this theorem, the load R_L will receive maximum power when, $R_L = R_{th}$

3. Equipment:

- ⇒ Digital Multimeter
- ⇒ DC Power supply adjusted to 20V DC
- ⇒ Resistor and Potentiometer
- ⇒ Wires

4. Circuit Diagram:

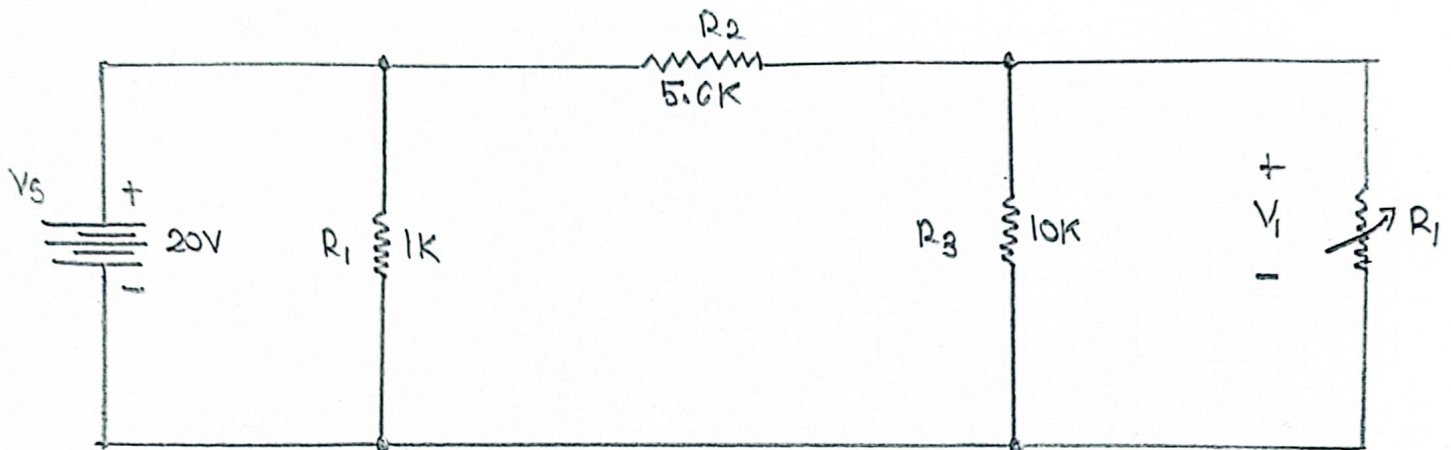


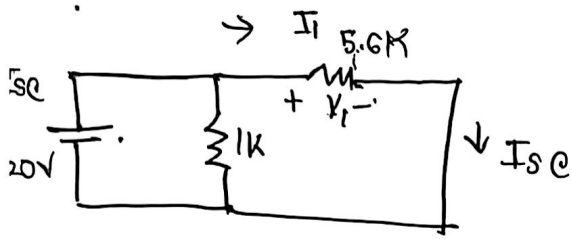
Fig 6: Circuit for verifying maximum power transfer theorem

5. Data Table:

R_L (k Ω)	V_1 (V)	$P_1 = V_1^2 / R_L$ (mW)
0.98	2.79	7.773 7.98
2.16	4.88	11.01
3.27	6.18	11.68
3.55 (R_{th})	6.45	11.70 (Maximum)
4.67	7.32	11.49
5.51	7.84	11.16
9.94	9.50	9.08

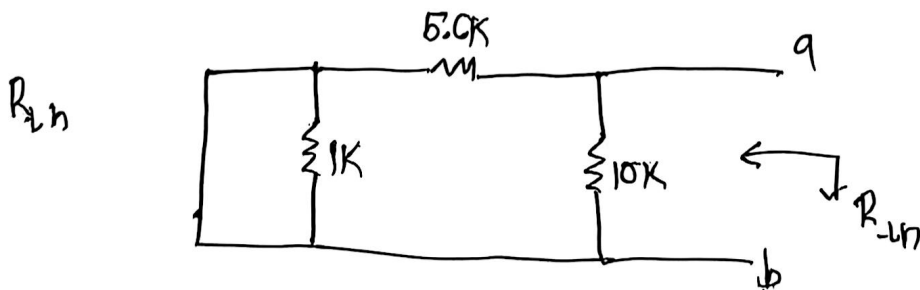
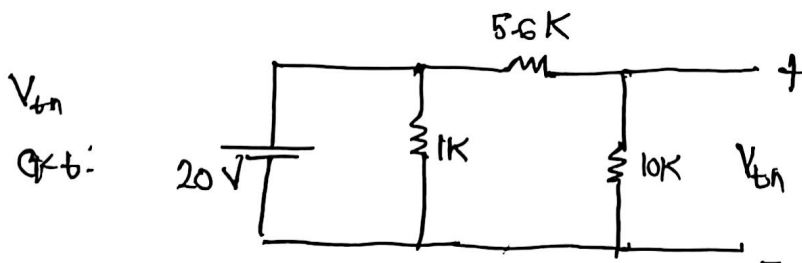
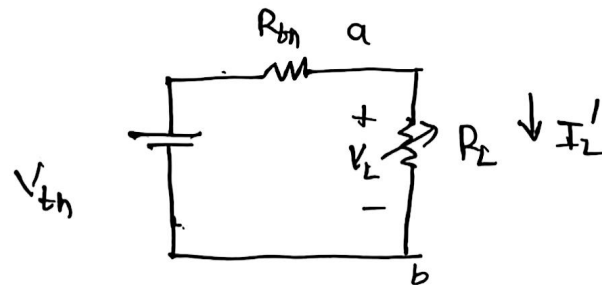
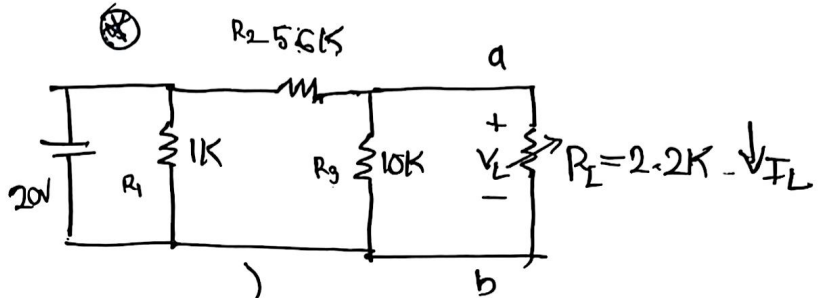
EEF101/L

Exp



$$I_{sc} = I_1 = \frac{V_1}{5.6K}$$

$$R_{th} = \frac{V_{th}}{I_{sc}}$$



$$R_1 = 0.986 \text{ k}$$

$$R_2 = 5.54 \text{ k}$$

$$R_3 = 10.01 \text{ k}$$

$$R_L = 2.16 \text{ k}$$

$$V = 20.1 \text{ V}$$

$$V_L = 4.99$$

$$I_L = 2.3101 \text{ mA}$$

$$V_{th} = 3.628$$

$$3.628$$

$$V_{th} = 12.9 \text{ V}$$

$$R_{th} = 3.555$$

$$V_L' = 4.975 \text{ V}$$

$$R_L = 2.16 \text{ k}$$

$$I_L' = \frac{V_L'}{R_L} = \frac{4.975}{2.16}$$

$$= \cancel{2.023 \text{ mA}} \\ = 2.301 \text{ mA}$$

$$V_{th} = 12.91 \text{ V}$$

$$V_1 = \cancel{17.12 \text{ V}} 20.1 \text{ V}$$

$$I_{sc} = \frac{V_1}{5.54} = \frac{3.628 \text{ mA}}{\cancel{3.057 \text{ mA}}}$$

$$R_{th} = \frac{V_{th}}{I_{sc}} = \frac{12.100}{3.628}$$

$$= 3.365 \text{ k}\Omega$$

$$R_{th} = 3.555 \text{ k}\Omega \text{ (direct)}$$

Data Table

Table 1: Data for original circuit

R_1 (K Ω)	V_{I_0} (V)	I_{I_0} (mA)
2.16	4.995	2.29

Table 2: Data for V_{tn}

V_{tn}
12.19

Table 3: Data for R_{tn}

Indirect:

R_2	V_1	$I_1 = V_1/R_2$	$I_{sc} = I_1$	$R_{tn} = V_{tn}/I_{sc}$
5.54	17.12 20.1	3.087 3.628	3.087 3.628	3.958 3.558

Direct:

R_{tn}
3.0566

Table 4: Data for eq^v circuit

V_{ie}	I_e
4.87 4.95	2.25 2.22

$$R_1 = 3.27$$

$$R_2 = 5.51$$

$$R_3 = 9.94$$

$$R_4 = 0.98$$


30/06/26

Data Table

R_1 (Ω)	V_{L1} (V)	$P_1 = V_{L1}^2 / R_1$ (mW)
0.98	2.76	7.773
2.16	4.81	0.969
3.27	6.09	1.756
4.67 3.55	6.24	10.96
4.67	7.24	3.012
5.51 5.51	7.75	3.917
9.94	9.37	10.54